

Homework (Habanero)

- Q1.** In a medical study, the dosage of a new drug is 243 milligrams. What is the value of the digit 4?
(A) 400
(B) 40
(C) 4
(D) 4000
(E) 0.4
- Q2.** A biologist counts 4567 cells in a sample. Rounded to the nearest hundred, how many cells are there?
(A) 4500
(B) 4600
(C) 5000
(D) 4000
(E) 46000
- Q3.** What is the expanded form of 945?
(A) $900 + 40 + 5$
(B) $900 + 40 + 50$
(C) $900 + 4 + 5$
(D) $900 + 45$
(E) $900 + 9 + 5$
- Q4.** A pharmaceutical company produces 1200 pills per hour. What is the standard form of this number?
(A) 1.2×10^3
(B) 1.2×10^2
(C) 12×10^2
(D) 12×10^3
(E) 0.12×10^3
- Q5.** In a medical trial, the success rate is 87.34%. Rounded to the nearest tenth, what is this percentage?
(A) 87.3%
(B) 87.4%
(C) 87%
(D) 88%
(E) 0.8734%
- Q6.** A doctor prescribes a 0.542-gram dose of medication. What is this value rounded to the nearest hundredth?
(A) 0.54
(B) 0.55
(C) 0.5
(D) 0.56
(E) 0.57
- Q7.** The average height of a population is 175.8 centimeters. Rounded to the nearest whole number, what is this height?
(A) 175
(B) 176
(C) 178
(D) 180
(E) 185
- Q8.** A medical device measures blood pressure in 125.2 millimeters of mercury. What is the value of the digit 5?
(A) 50
(B) 500
(C) 5
(D) 0.5
(E) 5000

Open Ended Questions (FRQ)

- Q9.** A hospital has 354 beds. If the hospital is currently at 87% capacity, how many beds are occupied? Show your work and explain your reasoning.
- Q10.** The number of patients in a hospital's emergency room can be modeled by the equation $y = 25x + 150$, where y is the number of patients and x is the number of hours after the hospital opens. If the hospital opens at 8 AM, how many patients will there be at 2 PM? Use place value to help you solve the problem.

Chef's Tips

- Tip 1: Encourage students to read each question carefully and identify the place value concept being tested.
- Tip 2: Remind students to show their work and explain their reasoning for free-response questions.
- Tip 3: Consider providing additional support for students who struggle with place value concepts, such as visual aids or extra practice problems.